

□

**Theorem 4.2.** *We have  $\theta_{LC_n}(X) \neq 0$  if and only if  $n < 8$ .*

*Proof.* By Lemma 3.7, we have

$$\tilde{L}_{a_1a_5} = \{a_1a_5, a_1a_4a_5, a_na_1a_5, a_na_1a_4a_5\}$$

and

$$\tilde{R}_{a_1a_5} = \{a_1a_5, a_1a_2a_5, a_1a_5a_6, a_1a_2a_5a_6\}.$$

For  $n \geq 8$ , we have  $|\tilde{L}_{a_1a_5}| = |\tilde{R}_{a_1a_5}| = 4$ . By Lemma 5.8 in [12], we compute the submatrix  $\tilde{L}_{a_1a_5} \times \tilde{R}_{a_1a_5}$ , extracted from the Cayley table  $(S, \star)$ , as follows:

|                | $a_1a_5$       | $a_1a_2a_5$ | $a_1a_5a_6$    | $a_1a_2a_5a_6$ |
|----------------|----------------|-------------|----------------|----------------|
| $a_1a_5$       | $a_1a_5$       | $a_1a_2a_5$ | $a_1a_5a_6$    | $a_1a_2a_5a_6$ |
| $a_1a_4a_5$    | $a_1a_4a_5$    | .           | .              | .              |
| $a_5a_na_1$    | $a_5a_na_1$    | .           | $a_na_1a_5a_6$ | .              |
| $a_4a_5a_na_1$ | $a_4a_5a_na_1$ | .           | .              | .              |

where the zero is represented with a dot. To compute the entries the table, at position  $(s, t)$ , we only need to verify that

$$(st)^+ = s^+ \quad \text{and} \quad (st)^* = t^*.$$

Since its determinant is zero, this is sufficient to show that the determinant of  $\theta_{LC_n}(X)$  is zero, for every  $n \geq 8$ .

It is evident that the determinant of the monoid  $LC_1$  is non-zero. The Cayley table of  $LC_2$  is as follows:

|       | $a_1$ | $a_2$ | 1     |
|-------|-------|-------|-------|
| $a_1$ | $a_1$ | .     | $a_1$ |
| $a_2$ | .     | $a_2$ | $a_2$ |
| 1     | $a_1$ | $a_2$ | 1     |

which indicates that the determinant is non-zero.

Now, suppose that  $3 \leq n \leq 7$ .

Let  $e$  be an idempotent of  $LC_n$  and let  $W_1W_2 \cdots W_{m_e}$  denote the canonical form of  $e$  over  $A$ . As  $n \leq 7$  and there is a two-letter distance between every neighboring segments of  $W$ , we have  $m_e \leq 2$ .

By Lemma 4.1, since  $n \leq 7$ , we have

$$l_1 = |\{W_i \mid |W_i| = 1, \text{ and } d(W_{i-1}, W_i) \geq 3\}| \leq 1.$$

Thus, it follows that  $1 \leq |\tilde{L}_e| \leq 2$ .

Then, we have the following cases:

$|\tilde{L}_e| = 1$  and  $m_e = 1$ :

In this case,  $e$  is an idempotent with a single segment of length greater than one. Thus, the submatrix  $\tilde{L}_e \times \tilde{R}_e$  contains only a single entity, which is  $e$ , and its determinant is non-zero.

$|\tilde{L}_e| = 1$  and  $m_e = 2$ :